rAIden

99

pg_crash

PGCONF.EU 2026 · VALENCIA · LIGHTNING TALK

ROUND 1...

FIGHT!





PG_CRASH VS RAIDEN

THE AI DBA THAT FIGHTS BACK AND GETS STRONGER · @KLOBA

rAIden

99

pg_crash

CHOOSE YOUR FIGHTER



rAIden

AUTONOMOUS ON-CALL DBA

WAKES UP IN	24 S
RUNBOOKS	11
GUARDRAIL TESTS	121
SLEEP NEEDED	0 H

SPECIAL MOVE: ROOT CAUSE + 1 RESTART

VS



PG_CRASH

CHAOS-MONKEY EXTENSION

SIGNAL EVERY	4 S
TARGET	RANDOM PID
SIGNALS	9 · 15 · 3
MERCY	NONE

SPECIAL MOVE: SIGKILL, RANDOM TARGET

PG_CRASH IS CYBERTEC'S OPEN-SOURCE CHAOS EXTENSION · RAIDEN IS THIS TALK

rAIden 99 pg_crash

HOW rAIden FIGHTS

1 WATCH

tails the Postgres jsonlog 24/7; a pg_isready heartbeat catches silent outages

↓ → P

2 TRIAGE

classifies lines, debounces storms: 20 crash lines = 1 incident

↓ ↓ K

3 FIGHT

tool-calling loop · GPT-5.5 · 11 skills (runbooks) · bash, files, probes

← → P

4 VERIFY

the ledger oracle: every commit the app was told OK about must still be there

↑ ↑ K

5 EVOLVE

writes a new check, a learned skill and lessons. Stronger next round

↓ ↑ P

THE REFEREE: GUARDRAILS

- no pkill
- no pg_resetwal
- never fsync=off
- no DROP on app data
- no network
- restart budget 6/h
- every call audited
- writes confined
- 121/121 tests pass

THE AI ASKED FOR BIGGER HAMMERS

```
# real refusals from the first full run
DENIED [sudo-scope]
  sudo pg_ctlcluster 18 main stop -m immediate
DENIED [pg-ctl-prod]
  sudo -u postgres pg_ctl … stop -m immediate
DENIED [systemctl-verb]
  systemctl kill --signal=KILL postgresql@18-main
```

CUSTOM HARNESS: ~3,000 LINES OF PYTHON, NOT AN SDK · GPT-5.5 ON AZURE OPENAI · NO API KEYS

rAlden

74

pg_crash

ROUND 1 · PG_CRASH STORM

T+0 s	a “temporary” experiment lands in conf.d
T+24 s	rAlden wakes: 1 incident, not 20 log lines
T+28 s	crash worker found in shared_preload_libraries
T+96 s	oracle green: 0 commits lost



acknowledged commits per 10 s · seconds around the attack

FINISH HIM!

A CRASH EVERY ~4 S. THE CULPRIT WAS IN THE CONFIG ALL ALONG

rAlden

58

pg_crash

FATALITY

```
# the real Postgres log from the lab VM
LOG:    client backend (PID 180019) was terminated by signal 9: Killed
LOG:    terminating any other active server processes
FATAL:  terminating connection due to administrator command
LOG:    all server processes terminated; reinitializing
FATAL:  the database system is in recovery mode
```

```
# conf.d/50-experiment.conf
- shared_preload_libraries = 'pg_stat_statements,pg_crash'
+ shared_preload_libraries = 'pg_stat_statements'
# the file was backed up first
# validated with postgres -C, then ONE restart
```

`1 restart (budget 6/h)` `96 s to a green oracle` `0 commits lost`

The Postgres log already prints FATAL:.
rAlden is just the first thing that *reads* it at 3 a.m.

FINISHING MOVE: REMOVE THE PRELOAD LIBRARY, RESTART ONCE, VERIFY

rAIden

81

pg_crash

ROUND 2 · TOAST

TOASTY!

4 rows damaged

rAIden: PITR side-clone → verified payloads → keyed UPDATE → amcheck clean

16 s to the clue · 90 s to a green oracle

The Oversized-Attribute Storage Technique

A FAT VALUE LIVES OUT-OF-LINE, IN PG_TOAST CHUNKS

0	×	2	3	4

```
ERROR:  unexpected chunk number 2 (expected 1)
        for toast value 756443 in pg_toast_706021
```



FOUR CHUNKS EATEN. RAIDEN BROUGHT THEM BACK FROM A TIME MACHINE

rAIden

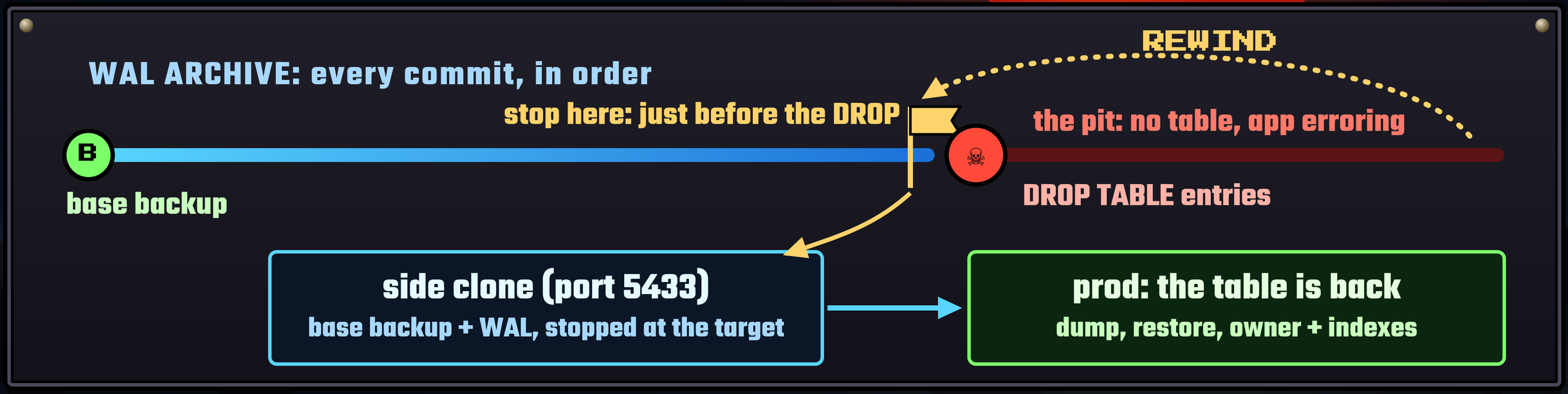
42

pg_crash

ROUND 3 · THE PIT

THE PITR

POINT-IN-TIME RECOVERY



xid found with pg_waldump	22 s to the clue	86 s to a green oracle	0 commits lost

THE PIT IS ONLY A FATALITY IF YOU HAVE NO BACKUPS. AND A PAGER

rAIden

pg_crash

AFTER THE FIGHT

LEVEL UP

rAIden wrote itself 8 new moves during the fight.
one of them defused a time bomb

UNLOCKED MOVES (SELF-WRITTEN)

```
+ learned-pg-crash-preload
+ learned-pg-surgery-toast
+ learned-protected-table-drop-pitr
+ 5 patrol checks (checks.d/85-89)
```

+ **lessons.md**: read before the next fight
+ every artifact is validated: broken ones are repaired or quarantined

REMATCH: THE TIME BOMB

T+0 s	experiment file back in conf.d + reload; restart later
T+4 s	rAIden's own check: the next restart would load pg_crash
T+28 s	file fixed, config reloaded: no restart needed
T+300 s	the scheduled restart fires: nothing happens

`0 crashes` `0 restarts by rAIden` `0 commits lost`

EVERY FIGHT LEAVES A NEW CHECK, A NEW RUNBOOK, A NEW LESSON


rAlden
pg_crash
FLAWLESS VICTORY
51,769
ACKNOWLEDGED COMMITS
0
LOST · 0 CORRUPT · 0 UNREADABLE
5/5
ROUNDS WON
OK
PG_AMCHECK
THE REFEREE: A CLIENT-SIDE LEDGER OF EVERY COMMIT THE APP WAS TOLD OK FOR

rAIden

09

pg_crash

CONTINUE?

INSERT COIN ▸

github.com/kloba/rAIden

- pg_crash by CYBERTEC: open-source chaos, used with love and fear
- the evolve loop: ideas from Hermes Agent, ECC, nanobot
- built and fought on Azure, deleted after the fight
- oracle, guardrails and every number here: in the repo



SCAN TO REMATCH

THANK YOU · NO DBAS WERE HARMED (THEY GOT SOME SLEEP)